

Southeastern Section - 68th Annual Meeting - 2019

Paper No. 16-7

Presentation Time: 1:00 PM-5:00 PM

STRUCTURAL MAPPING OF UPPER PLATE DILATING FRACTURES ON THE ISLAND OF MAKRYKYTHERA, GREECE AT THE LOCAL PORT USING DRONE IMAGING AND STATISTICAL ANALYSIS

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Makrykythera island contains the port to Kythera Island, a touristic attraction at the Aegean Sea and also an active tectonic area exposing the upper carbonate plate above the sea level. This southwestern part of the active tectonic forearc ridge of the current Hellenic Arc has shown multiple extensional stages that took place since Early Miocene and they are reflected in brittle and brittle-ductile structures, and it has shown that it is capable of strong earthquakes in a major touristic area. The area is in high-risk of natural hazards and reactivation of fractures may induce deterioration of the infrastructure. New structural observations have been made with drone imaging to produce a highly detailed map of the region to examine joint distribution. Using a drone flying at a height of 80 meters 1,546 images were collected and produced a digital surface model of 82 million points at a resolution of 4.13 centimeters per pixel to be used for structural mapping. Post-processing of the images was done in PIX4D software. Images were acquired with an autonomous drone, DJI Matrice M100, with a Zenmuse X3 camera. 899 joints were delineated inland and in the sea where the water was clear and shallow enough to obtain a clear identification of a fracture. The mapping of joints was done in Google Earth and the geodatabase was constructed in ArcMap. Statistical analysis was conducted in ArcMap with the geostatistical analyst tool, and an open-source Stereonet software to produce rose-diagrams. 854 gash fractures show a sum of 18,100 meters in length and infer an arc-normal and arc-parallel extension around the island and close to the port. The remaining 45 joints were determined to be fissure veins that show a total of 1,002 meters in length, and infer a consistent arc-parallel extension located far away from the port. We infer that multiple extensional stages facilitated the expansion of the arc and driven by the rollback of the African subducting plate provoke the occurrence of those fractures. Mapping of such fractures may predict future infrastructure failure and communication disruption with the mainland. Consideration of a high-resolution joint distribution map prior to any construction may mitigate future natural hazards.

Session No. 16--Booth# 7

[T12. Natural Hazards Prediction \(Posters\)](#)

Thursday, 28 March 2019: 1:00 PM-5:00 PM

[Carolina Ballroom \(Francis Marion Hotel\)](#)

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